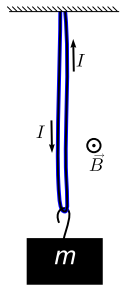
Problem 7. Lifting by current *(7 points)*



Consider a loop of freely deformable conducting wire with insulation of length $2l$, the two ends of which are fixed (permanently) to the ceiling. A load of mass m is fixed to the middle of the wire (the mass of the wire is negligible). There is also a horizontal magnetic field of induction B; free fall acceleration is g. A current I is lead through the wire. Neglect the field induced by the wires.

i. (2 pts) Sketch the shape of the wire.

ii. (2 pts) What is the maximal height by which the load can

U^{238}	Th^{234}	Pa^{234}	U^{234}	Th^{230}	Ra^{226}	Rn^{222}	Po^{2}
4.27	0.27	2.27	4.86	4.77	4.87	5.59	6.
17.15	6.32	4.38	12.89	12.38	10.70	5.52	2.

be lifted in such a way (increasing the current, if necessary)?

iii. (2 pts) Write an equation from which it is possible to determine the lifting height Δh.

iv. (1 pt) How large current I_0 is needed to lift the load by $\Delta h_0 = l(1 - \frac{3}{\pi})$?